

IFY STATISTICS | READ THE SYMBOLS, THEN CHOOSE THE MEASURE

01 WHOLE GROUP OR SAMPLE?

We want to study **all 200 gym members** (the population), but measure **12 members** (a sample). Their average and spread help us estimate the results for all 200.

μ - mu ("mew")	Population mean: average of the whole population.
$\bar{x}$ - x-bar	Sample mean: average of the sample; estimates μ .
σ - sigma	Population standard deviation (SD): spread around μ , in the original units.
s - letter s ("ess")	Sample standard deviation: spread from the sample; estimates σ .
σ^2 - sigma squared	Population variance: average squared difference from μ .
s^2 - s squared	Sample variance: squared differences from $\bar{x}$, added and divided by $n - 1$.
N / n	Capital N: population count. Small n: sample count.
x_i - x sub i	One observed value. The small i labels which observation.
Σ - capital sigma	The sum sign: add all the indicated terms.

Do not mix up these two symbols

Σ = **capital sigma:** an instruction to add.

σ = **lowercase sigma:** population standard deviation.

F1 MEAN: ADD, THEN DIVIDE

$$\mu = \frac{\sum x_i}{N} \quad \bar{x} = \frac{\sum x_i}{n}$$

Read both: mean = total of the values $\div$ number of values.

A **parameter** describes a population. A **statistic** is calculated from a sample. The sample result describes those measured and helps estimate the larger group.

02 MEAN, MEDIAN AND MODE

Mean	Add all values; divide by the count.
Median	Sort first. Read the middle value; for an even count, average the two middle values.
Mode	Most frequent value or category; there may be more than one mode, or none.

Example: 4, 6, 6, 8, 11, 13

Mean: $48 \div 6 = 8$.

Median: $(6 + 8) \div 2 = 7$.

Mode: 6.

F2 MEDIAN POSITION

$$\text{position} = \frac{n+1}{2}$$

Here n is the **count of ordered values**.

For 1,000 values, average the **values at positions 500 and 501**. A position is not a data value.

03 QUARTILES AND THE MIDDLE HALF

Q1 - Q one	Lower quartile: start of the middle half.
Q2 - Q two	Second quartile: the median.
Q3 - Q three	Upper quartile: end of the middle half.
IQR	Interquartile range: width of the middle half; $Q_3 - Q_1$.
Range	Maximum (largest) – minimum (smallest).

F3 QUARTILE POSITIONS

$$Q_1 : \frac{n+1}{4} \quad Q_3 : \frac{3(n+1)}{4}$$

Interpolation: use part of the gap between two ordered values. At position 3.25, with 3rd value 8 and 4th value 12:

$$8 + 0.25 \times (12 - 8) = 9.$$

Five-number summary: minimum, Q_1 , median, Q_3 , maximum.

Resistant: less affected by extreme values. Median and IQR can still change.

IFY STATISTICS | UNDERSTAND SPREAD AND DISTRIBUTION SHAPE

04 DEVIATION, VARIANCE AND SD

A **deviation** is one value minus the mean. If the value is 6 and the mean is 10, the deviation is **-4**: the value is 4 below the mean.

	DEVIATION	STANDARD DEVIATION
Describes	One value	All recorded values
Sign	Negative, zero or positive	Zero or positive
Meaning	Difference and direction	Overall spread around the mean

Small SD: values are close together around the mean.

Large SD: they are more spread out.

F4 CALCULATE VARIANCE AND SD

Population variance and standard deviation

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}, \quad \sigma = \sqrt{\sigma^2}$$

Sample variance and standard deviation

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}, \quad s = \sqrt{s^2}$$

Read $\Sigma(x_i - \text{mean})^2$: subtract the mean from each value, square each difference, then add.

2 means squared: multiply by itself.

$\sqrt{}$ means square root: reverses squaring.

Variance uses squared units (min^2); SD returns to the original units (min).

Follow one sample calculation

Delivery times (min): 20, 13, 4, 8, 10.

1. Mean = $55 \div 5 = 11$ min.

2. Deviations: 9, 2, -7, -3, -1.

3. Square and add: $81 + 4 + 49 + 9 + 1 = 144$. This total is the **sum of squares (SS)**.

4. Sample variance: $144 \div (5 - 1) = 36 \text{ min}^2$.

5. Sample SD: $\sqrt{36} = 6$ min.

Why $n - 1$? Using the sample's own mean tends to make variance calculated with n too small. The divisor $n - 1$ adjusts the estimate. We still use **every observation**.

05 COEFFICIENT OF VARIATION (CV)

CV compares SD with the size of the mean. It gives spread as a **percentage of the mean**. A lower CV means less relative variation.

$$\text{Population: } CV = \frac{\sigma}{\mu} \times 100\%$$

$$\text{Sample: } CV = \frac{s}{\bar{x}} \times 100\%$$

	MEAN (mL)	SD (mL)	CV
Small bottles	248.6	6.8	2.74%
Large bottles	742.3	18.9	2.55%

Small bottles have less spread in mL (**absolute variation**). Large bottles have less spread compared with their mean (**relative variation**).

Use CV when the comparison makes sense

The mean must be positive and away from zero. Zero should mean **none of the quantity**, as with 0 mL. Lower CV means relative consistency, not accuracy against a target.

06 SHAPE: SKEWNESS AND KURTOSIS

A **distribution** shows the pattern of values. Its **tails** contain values far from the centre.

Left skew



Symmetric



Right skew



Negative / left skew: long tail to smaller values.

Symmetric: both sides have similar shapes.

Positive / right skew: long tail to larger values.

Name skewness by the **long tail**, not the peak.

Normal distribution: a symmetric, bell-shaped reference.

Kurtosis: how strongly extreme values influence the distribution. Look beyond the peak.

Mesokurtic

Normal-reference level.

Leptokurtic

Higher kurtosis; stronger influence from extremes.

Platykurtic

Lower kurtosis; weaker influence from extremes.

Week 3 uses descriptions and graphs for kurtosis.